

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 2

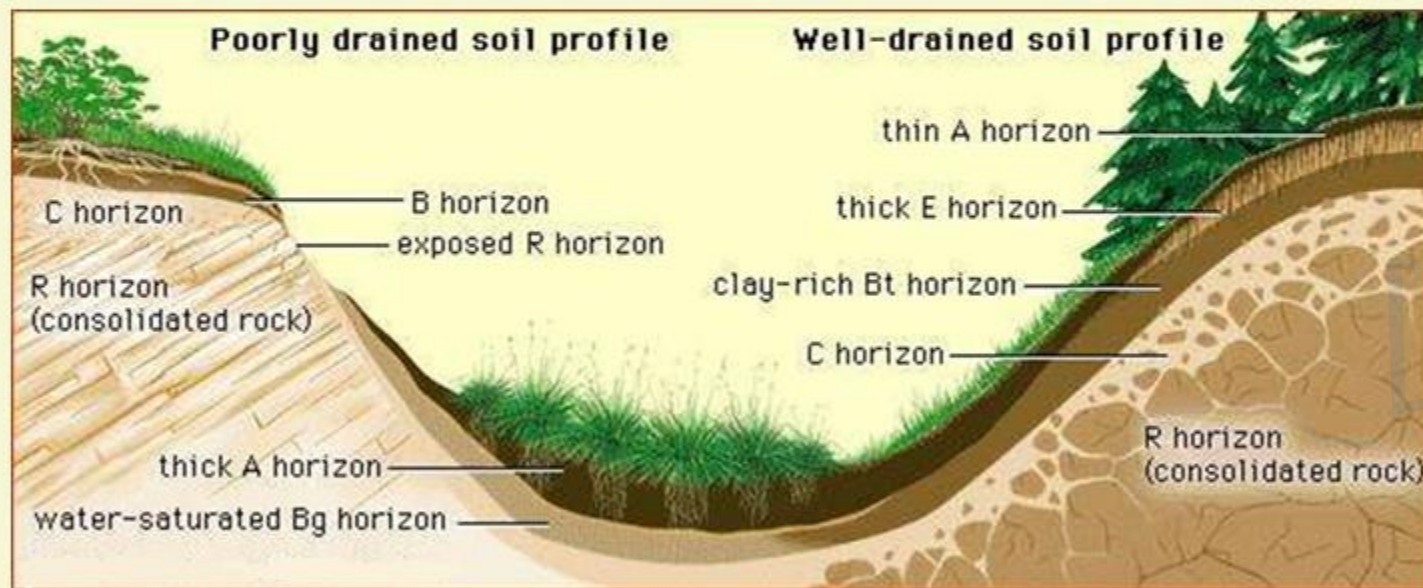
SOIL AND ITS CONTENTS

- **PROCESS OF SOIL FORMATION**
- **SOIL FORMING FACTORS**

SOIL-FORMING FACTORS

2

TOPOGRAPHY



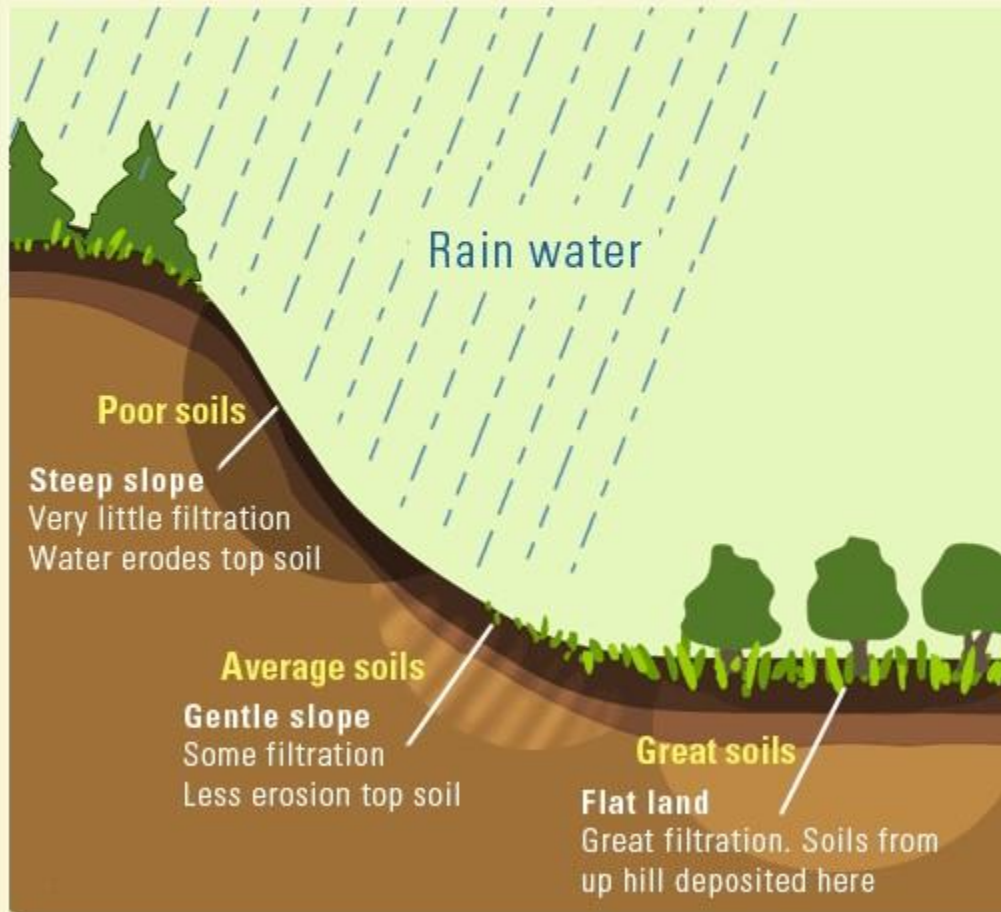
1 Topography like parent materials is another **passive control factor**.

2 The influence of topography is felt through **the amount of exposure** of a surface covered by parent materials to **sunlight and the amount of surface and sub-surface drainage** through the parent materials.

SOIL-FORMING FACTORS

2

TOPOGRAPHY



3 Soils will be thin on steep slopes and thick over flat upland areas.

4 Over gentle slopes where erosion is slow and percolation of water is good, soil formation is very favourable.

5 Soils over flat areas may develop a thick layer of clay with good accumulation of organic matter giving the soil dark color.

SOIL-FORMING FACTORS

3

CLIMATE



Climate is an **important active factor** in soil formation. The climatic elements involved in soil development are:

- i. **Moisture** in terms of its intensity, frequency and duration of precipitation-evaporation and humidity.
- ii. **Temperature** in terms of seasonal and diurnal variations.

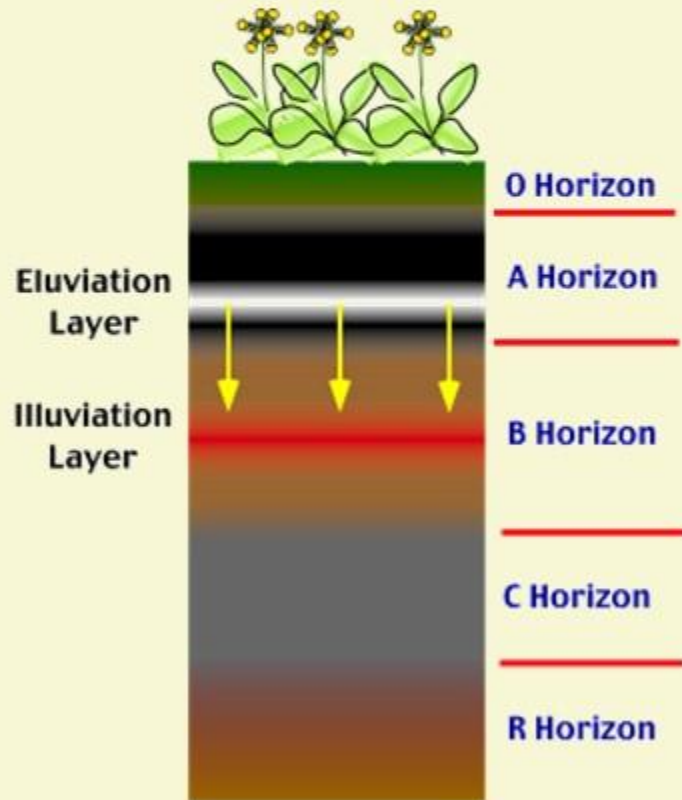
SOIL-FORMING FACTORS

3

CLIMATE

a

MOISTURE



1

Precipitation gives soil its moisture content which makes the chemical and biological activities possible.

2

Excess of water helps in the downward transportation of soil components through the soil (**eluviation**) and deposits the same down below (**illuviation**).

3

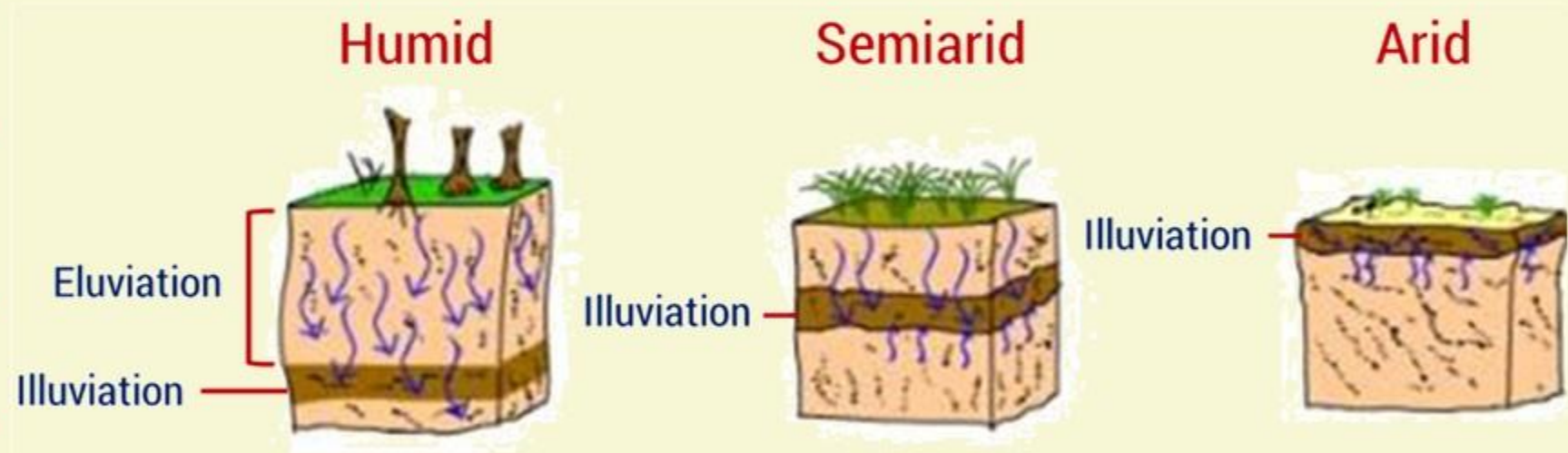
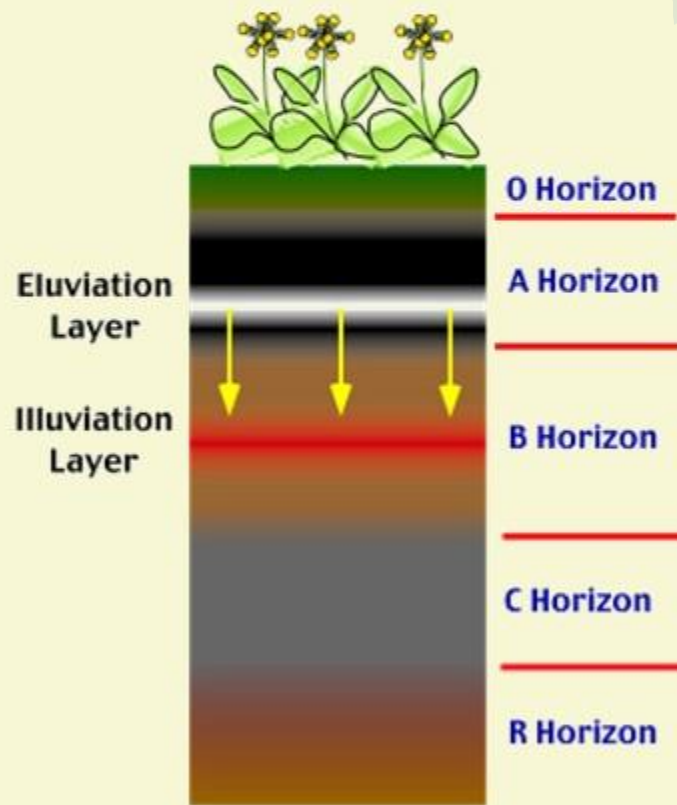
In climates like wet equatorial rainy areas with high rainfall, not only calcium, sodium, magnesium, potassium etc. but also a major part of silica is removed from the soil.

SOIL-FORMING FACTORS

3

CLIMATE

ELLUVIATION VS ILLUVIATION



SOIL-FORMING FACTORS

3

CLIMATE

a

MOISTURE

- 4 Removal of silica from the soil is known as **desilication**. In dry climates, because of high temperature, evaporation exceeds precipitation



- 5 Hence ground water is brought up to the surface by **capillary action** and in the process the water evaporates leaving behind salts in the soil.

- 6 Such salts form into a crust in the soil known as **hardpans**.

- 7 In tropical climates and in areas with intermediate precipitation conditions, **calcium carbonate nodules (kanker)** are formed.

SOIL-FORMING FACTORS

3

CLIMATE

b

TEMPERATURE



1

Temperature acts in two ways:
increasing or reducing chemical and biological activity.

2

Chemical activity is increased in higher temperatures, reduced in cooler temperatures and stops in freezing conditions.

3

That is why, tropical soils with higher temperatures show deeper profiles and in the frozen tundra regions soils contain largely mechanically broken materials.

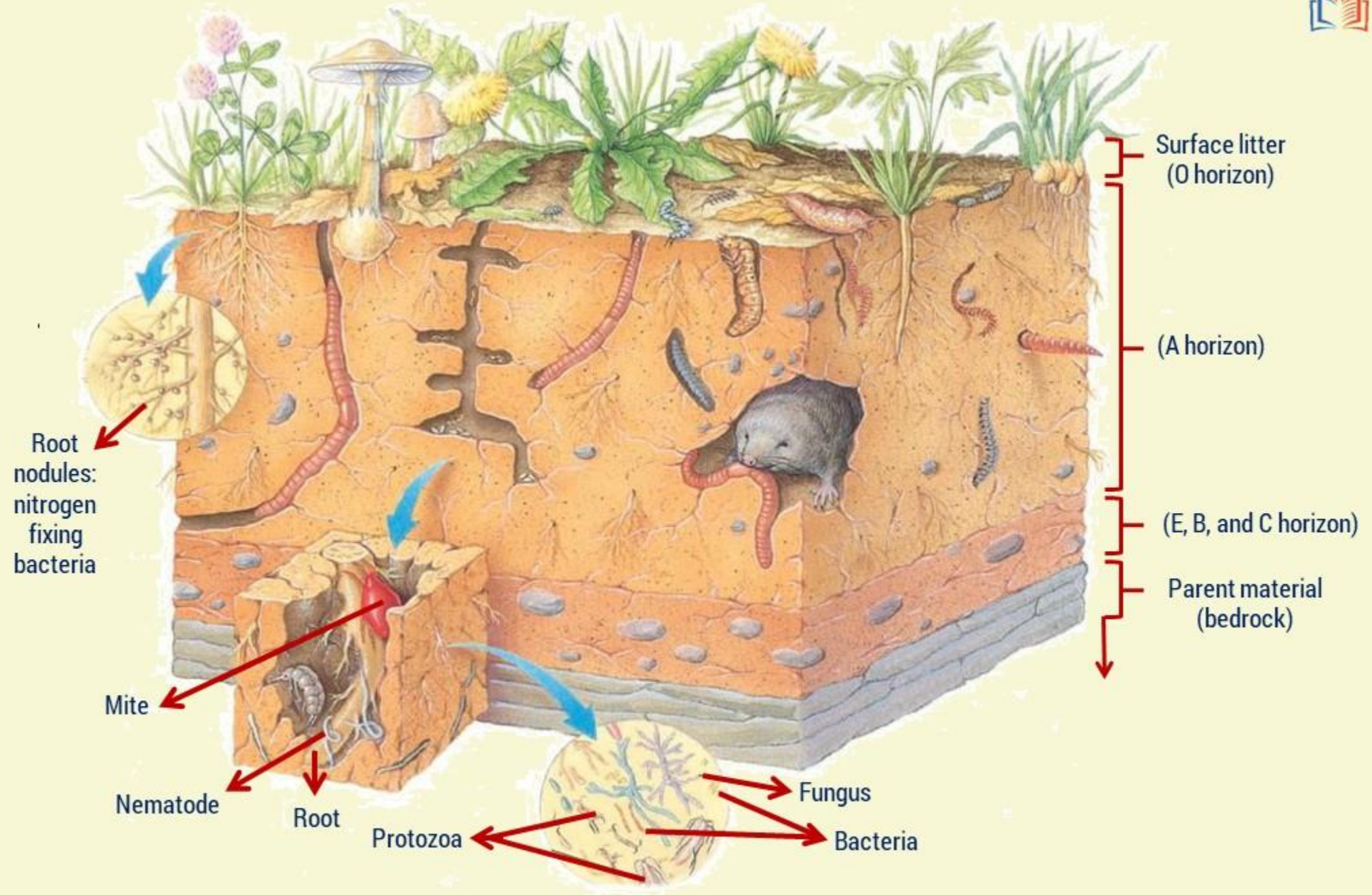
SOIL-FORMING FACTORS

4

BIOLOGICAL ACTIVITY



- 1 The vegetative cover and organisms that occupy the parent materials from the beginning and also at later stage help in adding organic matter, moisture retention, nitrogen etc.
- 2 Dead plants provide humus, the finely divided organic matter of the soil.
- 3 Some organic acids which form during humification aid in decomposing the minerals of the soil parent materials.



SOIL-FORMING FACTORS

4

BIOLOGICAL ACTIVITY

4 Intensity of bacterial activity shows up differences between soils of cold and warm climates.

5 Humus accumulates in cold climates as bacterial growth is slow. With undecomposed organic matter because of low bacterial activity, layers of peat develop in sub-arctic and tundra climates.

6 In humid tropical and equatorial climates, bacterial growth and action is intense and dead vegetation is rapidly oxidized leaving very low humus content in the soil.

7 Further, bacteria and other soil organisms take gaseous nitrogen from the air and convert it into a chemical form that can be used by plants. This process is known as Nitrogen Fixation.

SOIL-FORMING FACTORS

4

BIOLOGICAL ACTIVITY



8 Rhizobium, a type of bacteria, lives in the root nodules of leguminous plants and fixes nitrogen beneficial to the host plant.

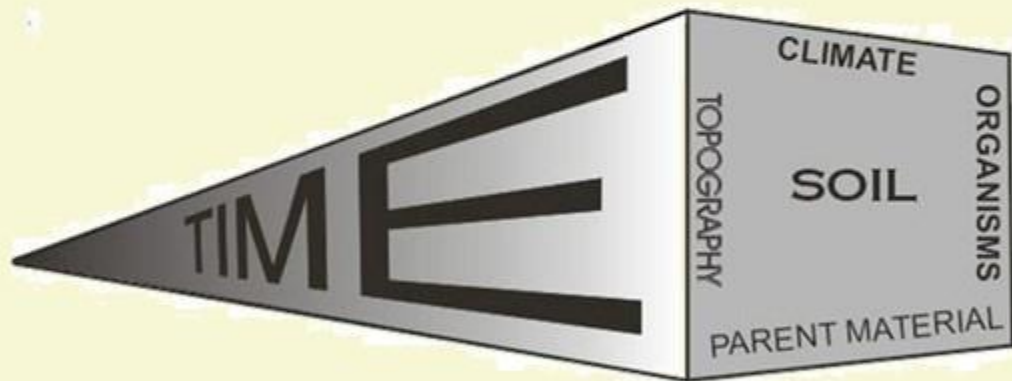
9 The influence of large animals like ants, termites, earthworms, rodents etc. is mechanical, but, it is nevertheless important in soil formation as they rework the soil up and down.

10 In case of earthworms, as they feed on soil, the texture and chemistry of the soil that comes out of their body changes.

SOIL-FORMING FACTORS

5

TIME



- 1 Time is the last important controlling factor in soil formation. The **length of time** the soil forming processes operate, determines **maturation of soils and profile development**.
- 2 A **soil becomes mature** when all soil-forming process act for a sufficiently long time developing a profile.
- 3 Soils developing from **recently deposited alluvium or glacial till** are considered **young** and they exhibit **no horizons** or only poorly developed horizons.
- 4 No specific length of time in absolute terms can be fixed for soils to develop and mature.

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